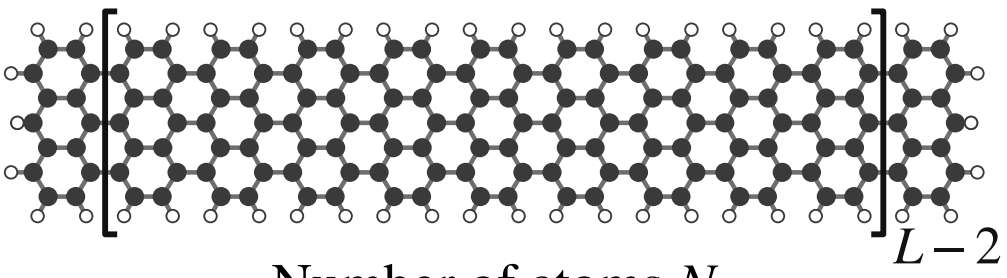
(a)



Number of atoms N

26 44 80 152 296 584 1,160 2,312 4,616 9,224

(b)

HOMO-LUMO gap (eV)

9 8 7 6 5 4

Tensor
RI-RS
RI-RS-Opt
aug-cc-pVDZ
aug-SZV-MOLOPT-ae-mini

3.971 eV

(c)

G_0W_0 cost (core hours)

10^6 10^5 10^4 10^3 10^2 10^1 10^0 10^{-1} 10^{-2} 10^{-3}

$L^{2.3}$ $L^{1.9}$ $L^{2.1}$ $L^{3.2}$ $L^{2.4}$

(d)

Cost of the steps (core hours)

10^6 10^5 10^4 10^3 10^2 10^1 10^0 10^{-1} 10^{-2} 10^{-3}

RI-RS-Opt total
Grid optimization $(L^{1.3})$
Z_{lP} $(L^{1.1})$
$\chi(i\tau)$ $(L^{1.7})$
$W(i\tau)$ $(L^{2.6})$
$\Sigma^x + \Sigma^c(i\tau)$ $(L^{1.7})$
QP $(L^{2.5})$

$L^{1.9}$

1 2 4 8 16 32 64 128 256 512

Number of strips L